

Data Intelligence Platform - Entity Relationship Diagram

Phase 3 deliverable (SOW 6). Sources: US Consumer Price Index (BLS) and Secured Overnight Financing Rate (NY Fed).

PK Primary key **FK** Foreign key **UQ** Part of a unique key **UQf** Unique where IsCurrent = 1

COLLECT

DataSource	collect
DataSourceId PK	TINYINT
Code UQ	VARCHAR(20)
Name	NVARCHAR(100)
Publisher	NVARCHAR(100)
ApiEndpoint	NVARCHAR(1000)
AccessMethod	VARCHAR(20)
HttpMethod	VARCHAR(6)
PublicationCadence	VARCHAR(20)
IsEnabled	BIT
2 rows: BLS_CPI, NYFED_SOFR	

CollectionRun	collect
CollectionRunId PK	BIGINT
DataSourceId FK UQ	TINYINT
ScheduledForUtc UQ	DATETIME2(0)
Attempt UQ	TINYINT
Status	VARCHAR(20)
StartedAtUtc	DATETIME2(3)
CompletedAtUtc	DATETIME2(3)
ObservationsInserted	INT
ObservationsRevised	INT
FailureCategory	VARCHAR(30)
one row per attempt (FR-2)	

RawPayload	collect
RawPayloadId PK	BIGINT
CollectionRunId FK	BIGINT
FetchAtUtc	DATETIME2(3)
ContentHash	BINARY(32)
SizeBytes	INT
CompressedContent	VARBINARY(MAX)
diagnostic; purge beyond ~90 days	

CORE

SeriesCategory	core
CategoryId PK	INT
ParentCategoryId FK	INT
Code UQ	NVARCHAR(100)
DisplayName	NVARCHAR(200)
SortOrder	SMALLINT
self-referencing: CPI item hierarchy	

Series	core
SeriesId PK	INT
DataSourceId FK UQ	TINYINT
SeriesCode UQ	NVARCHAR(100)
CategoryId FK	INT
FirstSeenRunId FK	BIGINT
Title	NVARCHAR(400)
Unit	NVARCHAR(60)
Frequency	VARCHAR(20)
SeasonalAdjustment	VARCHAR(24)
IsSourceAssignedCode	BIT
SourceFieldPath	NVARCHAR(200)
IsActive	BIT
one measured quantity; dedup anchor (FR-3)	

Observation	core
ObservationId PK	BIGINT
ReferenceDate PK UQ	DATE
SeriesId FK UQ	INT
RevisionNumber UQ	SMALLINT
IsCurrent UQf	BIT
SupersededAtUtc	DATETIME2(3)
PeriodType	VARCHAR(12)
Value	DECIMAL(28,8)
SourceAnnotation	VARCHAR(100)
CollectionRunId FK	BIGINT
CollectedAtUtc	DATETIME2(3)
RowHash	BINARY(32)
append-only fact; bi-temporal (FR-4, FR-6)	

RejectedObservation	core
RejectedObservationId PK	BIGINT
CollectionRunId FK	BIGINT
SeriesCode	NVARCHAR(100)
ReferenceDateText	NVARCHAR(50)
Reason	VARCHAR(30)
ReasonDetail	NVARCHAR(1000)
RejectedAtUtc	DATETIME2(3)
quarantine; early signal of schema drift	

SEC

AppUser	sec
UserId PK	INT
Email UQ	NVARCHAR(256)
DisplayName	NVARCHAR(150)
PasswordHash	NVARCHAR(500)
IsActive	BIT

UserRole	sec
UserId PK FK	INT
RoleId PK FK	TINYINT
GrantedAtUtc	DATETIME2(3)

Role	sec
RoleId PK	TINYINT
Name UQ	NVARCHAR(50)
Description	NVARCHAR(250)
Administrator / Analyst / Viewer	

AI

AssistantSession	ai
SessionId PK	UNIQUEIDENTIFIER
UserId FK	INT
StartedAtUtc	DATETIME2(3)

AssistantQuery	ai
AssistantQueryId PK	BIGINT
SessionId FK	UNIQUEIDENTIFIER
UserId FK	INT
QuestionText	NVARCHAR(2000)
GeneratedSql	NVARCHAR(MAX)
ValidationOutcome	VARCHAR(20)
WasExecuted	BIT
AnswerText	NVARCHAR(MAX)
AskedAtUtc	DATETIME2(3)
every generated statement logged (auditability)	

AssistantFeedback	ai
AssistantQueryId PK FK	BIGINT
IsHelpful	BIT
Comment	NVARCHAR(1000)